House Price Prediction Using Machine Learning

System Architecture Specification & Technical Documentation — Category: Predictive Regression & Machine Learning

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Repository: https://github.com/rakshapatil526/House-Price-Prediction

Status: Verified Repository

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1. Real Estate Valuation & Problem Statement

Real estate pricing is driven by complex non-linear interactions between geographic location, living area, architectural layout (bedrooms, bathrooms, floors), and property age. This machine learning application implements multi-variable regression to deliver transparent, data-driven property valuations for Bengaluru neighborhoods.

2. Feature Engineering & Column Transformation

Feature Category	Raw Inputs	Transformation Pipeline
Categorical Location	Whitefield, Indiranagar, Koramangala, etc.	OneHotEncoder(handle_unknown='ignore')
Continuous Dimensions	Area (sq.ft.), Property Age	StandardScaler normalization
Discrete Structural	Bedrooms (1-6), Bathrooms, Floors	StandardScaler + polynomial feature checks

3. Regression Model Benchmarking

• **Algorithms Benchmarked:** Linear Regression, Ridge/Lasso regularized regression, and Random Forest Regressor.

• **Performance:** Random Forest Regressor captures non-linear location premiums effectively, minimizing Root Mean Squared Error (RMSE) and achieving high R^2 valuation fidelity.

• **Persistent Log:** Evaluated quotes logged with timestamp, area, and predicted price (in Lakhs INR) to a persistent CSV database.

4. Application Stack

• **Stack:** Python 3.10+, Scikit-learn, Pandas, Streamlit, Matplotlib.

• **Status:** Fully verified from repository source code and model evaluation scripts.